ADVANCED PYTHON PROGRAMMING LAB ASSIGNMENT-07

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EXPLORING VIUSAL CAPITALIST - WORKING ON SUMMARIES,PIVOT TABLE

In [2]:
```
pip install pandas
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in ./.local/lib/python3.13/site-packages (3.0.5)
Requirement already satisfied: numpy>=1.26.0 in ./.local/lib/python3.13/site-packages (from pandas) (2.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/anaconda/lib/python3.13/site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: six>=1.5 in /opt/anaconda/lib/python3.13/site-packages (from python-dateutil>=2.8.2->
pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [10]:
```
import pandas as pd

#creating data
data = [
    [1, "Tampa", "USA", 36.1],
    [2, "Breda", "NLD", 29.3],
    [3, "Singapore", "SGP", 29.3],
    [4, "Oslo", "NOR", 28.8],
    [5, "Sydney", "AUS", 25.9],
    [6, "Vancouver", "CAN", 25.9],
    [7, "Montreal", "CAN", 25.5],
```

```
    [8, "Cambridge", "USA", 25.3],
    [9, "Xalapa", "MEX", 24.0],
    [10, "Durban", "ZAF", 23.7],
    [11, "Johannesburg", "ZAF", 23.6],
    [12, "Sacramento", "USA", 23.6],
    [13, "Frankfurt", "DEU", 21.5],
    [14, "Geneva", "CHE", 21.4],
    [15, "Veracruz", "MEX", 21.4],
    [16, "Amsterdam", "NLD", 20.6],
    [17, "Seattle", "USA", 20.0],
    [18, "Toronto", "CAN", 19.5],
    [19, "Miami", "USA", 19.4],
    [20, "Boston", "USA", 18.2],
    [21, "Tel Aviv", "ISR", 17.5],
    [22, "Guadalajara", "MEX", 17.3],
    [23, "León", "MEX", 16.5],
    [24, "Turin", "ITA", 16.2],
    [25, "Los Angeles", "USA", 15.2],
    [26, "Buenos Aires", "ARG", 14.5],
    [27, "Franca", "BRA", 13.7],
    [28, "New York", "USA", 13.5],
    [29, "Cape Town", "ZAF", 13.4],
    [30, "London", "GBR", 12.7]
]

#create data frame

df = pd.DataFrame(data, columns=["Rank", "City", "Country", "GVI (%)"])
df
```

Out[10]:

	Rank	City	Country	GVI (%)
0	1	Tampa	USA	36.1
1	2	Breda	NLD	29.3
2	3	Singapore	SGP	29.3
3	4	Oslo	NOR	28.8
4	5	Sydney	AUS	25.9
5	6	Vancouver	CAN	25.9
6	7	Montreal	CAN	25.5
7	8	Cambridge	USA	25.3
8	9	Xalapa	MEX	24.0
9	10	Durban	ZAF	23.7
10	11	Johannesburg	ZAF	23.6
11	12	Sacramento	USA	23.6
12	13	Frankfurt	DEU	21.5
13	14	Geneva	CHE	21.4
14	15	Veracruz	MEX	21.4
15	16	Amsterdam	NLD	20.6
16	17	Seattle	USA	20.0
17	18	Toronto	CAN	19.5
18	19	Miami	USA	19.4
19	20	Boston	USA	18.2
20	21	Tel Aviv	ISR	17.5
21	22	Guadalajara	MEX	17.3
22	23	León	MEX	16.5
23	24	Turin	ITA	16.2

	Rank	City	Country	GVI (%)
24	25	Los Angeles	USA	15.2
25	26	Buenos Aires	ARG	14.5
26	27	Franca	BRA	13.7
27	28	New York	USA	13.5
28	29	Cape Town	ZAF	13.4
29	30	London	GBR	12.7

In [13]:
```
# Find the average of GVI of "USA"

pivot_df=df.pivot_table(index="Country",values="GVI (%)",aggfunc=["mean","min","sum"])
pivot_df.columns = ["Average_GVI", "Minimum_GVI", "Total_Sum_GVI"]
pivot_df
```

Out[13]:

Country	Average_GVI	Minimum_GVI	Total_Sum_GVI
ARG	14.500000	14.5	14.5
AUS	25.900000	25.9	25.9
BRA	13.700000	13.7	13.7
CAN	23.633333	19.5	70.9
CHE	21.400000	21.4	21.4
DEU	21.500000	21.5	21.5
GBR	12.700000	12.7	12.7
ISR	17.500000	17.5	17.5
ITA	16.200000	16.2	16.2
MEX	19.800000	16.5	79.2
NLD	24.950000	20.6	49.9
NOR	28.800000	28.8	28.8
SGP	29.300000	29.3	29.3
USA	21.412500	13.5	171.3
ZAF	20.233333	13.4	60.7

1. Average of GVI of "USA"

In [15]:
```
average_gvi=pivot_df.loc["USA","Average_GVI"]
print("The Average GVI of USA is : ",average_gvi)
```

```
The Average GVI of USA is :  21.4125
```

2. Minimum of GVI of "NetherLands"

In [16]:
```
minimum_gvi=pivot_df.loc["NLD","Minimum_GVI"]
print("The minimum of GVI of Netherlands is : ",minimum_gvi)
```

```
The minimum of GVI of Netherlands is :  20.6
```

3. Sum of GVI of cities of "Italy"

In [17]:
```
sum_gvi=pivot_df.loc["ITA","Total_Sum_GVI"]
print("The sum of GVI of cities of italy : ",sum_gvi)
```

```
The sum of GVI of cities of italy :  16.2
```

4.Find the country which has maximum number of cities among top 30

In [24]:
```
city_pivot = df.pivot_table(
    index="Country",
    values="City",
    aggfunc="count"
)

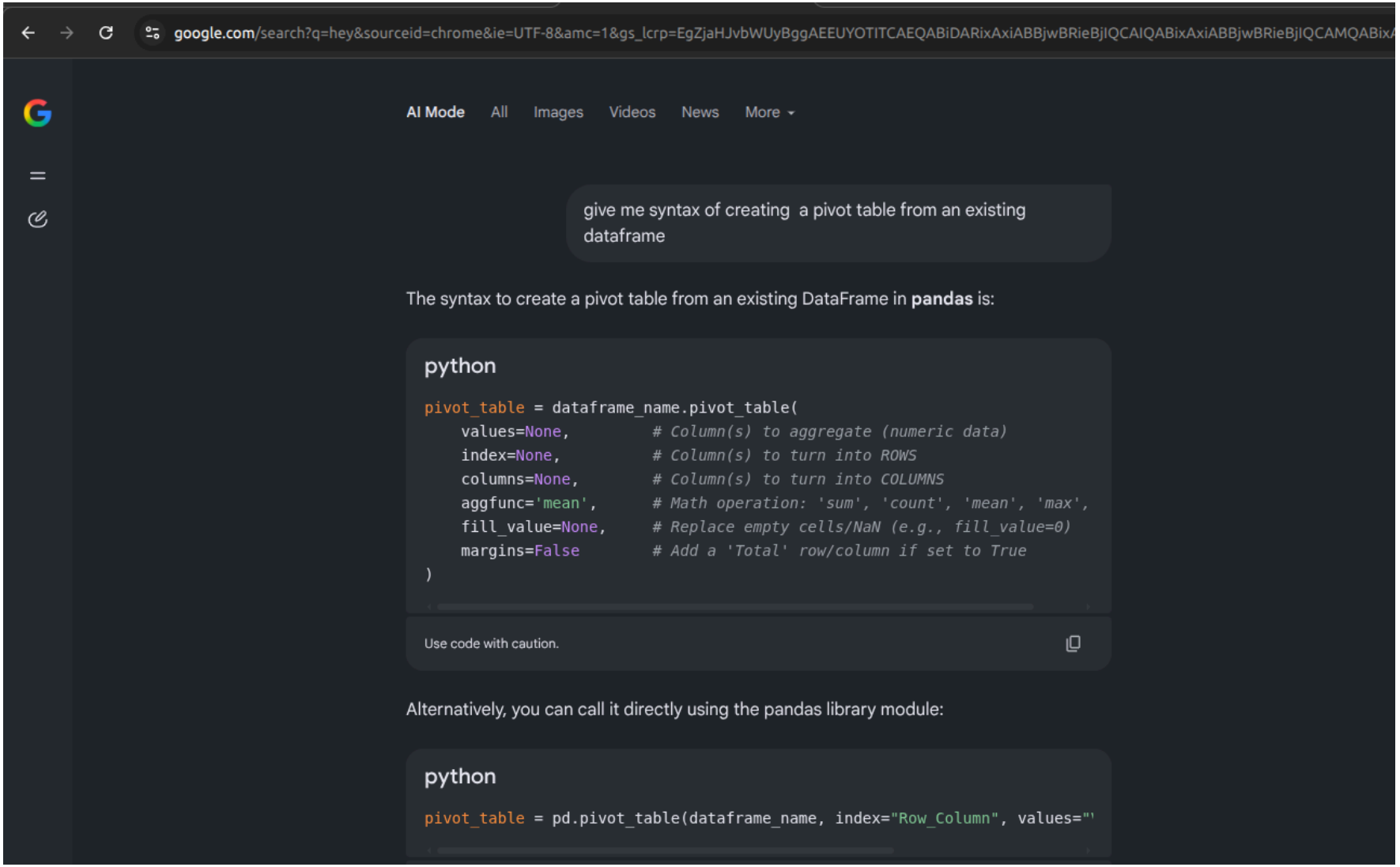
city_pivot.columns = ["City_Count"]

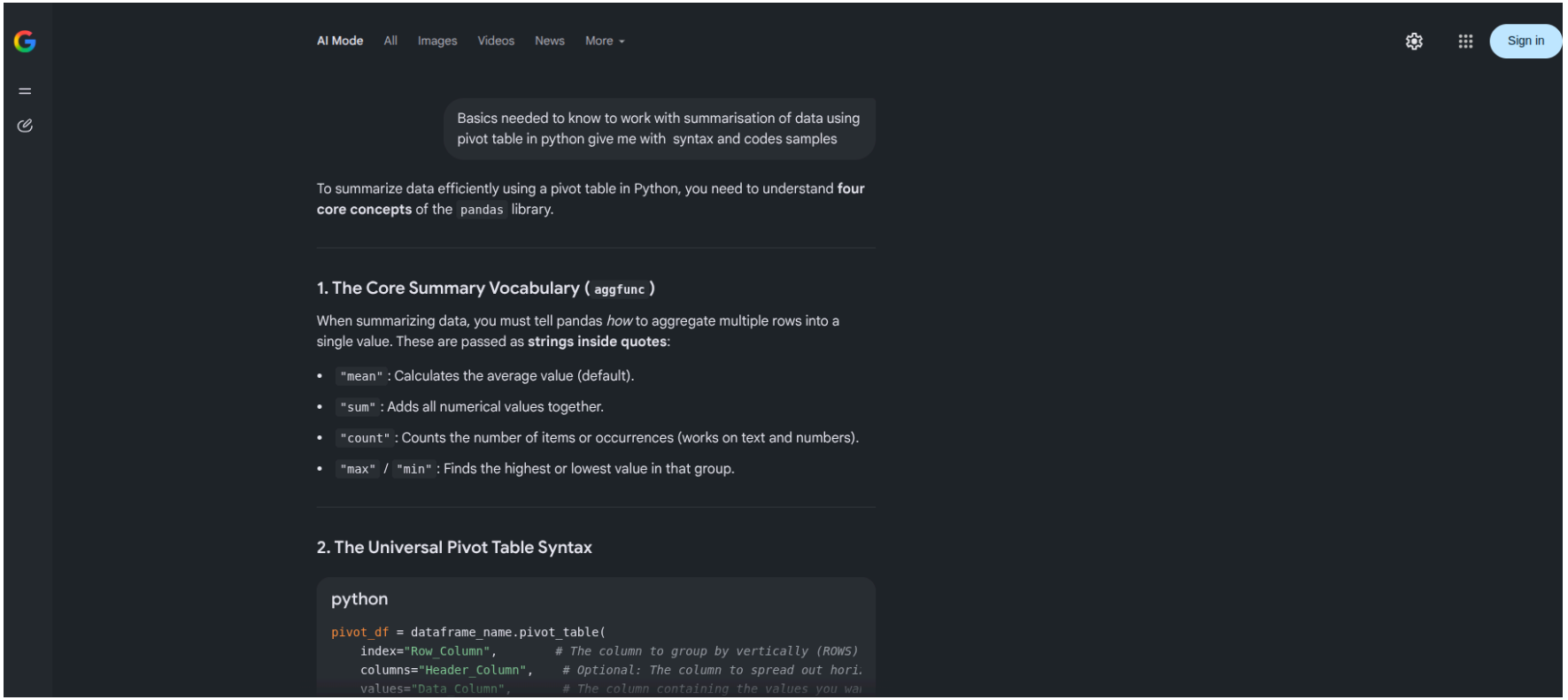
max_city_count= city_pivot["City_Count"].max()
top_country = city_pivot[city_pivot["City_Count"] == max_city_count].index[0]

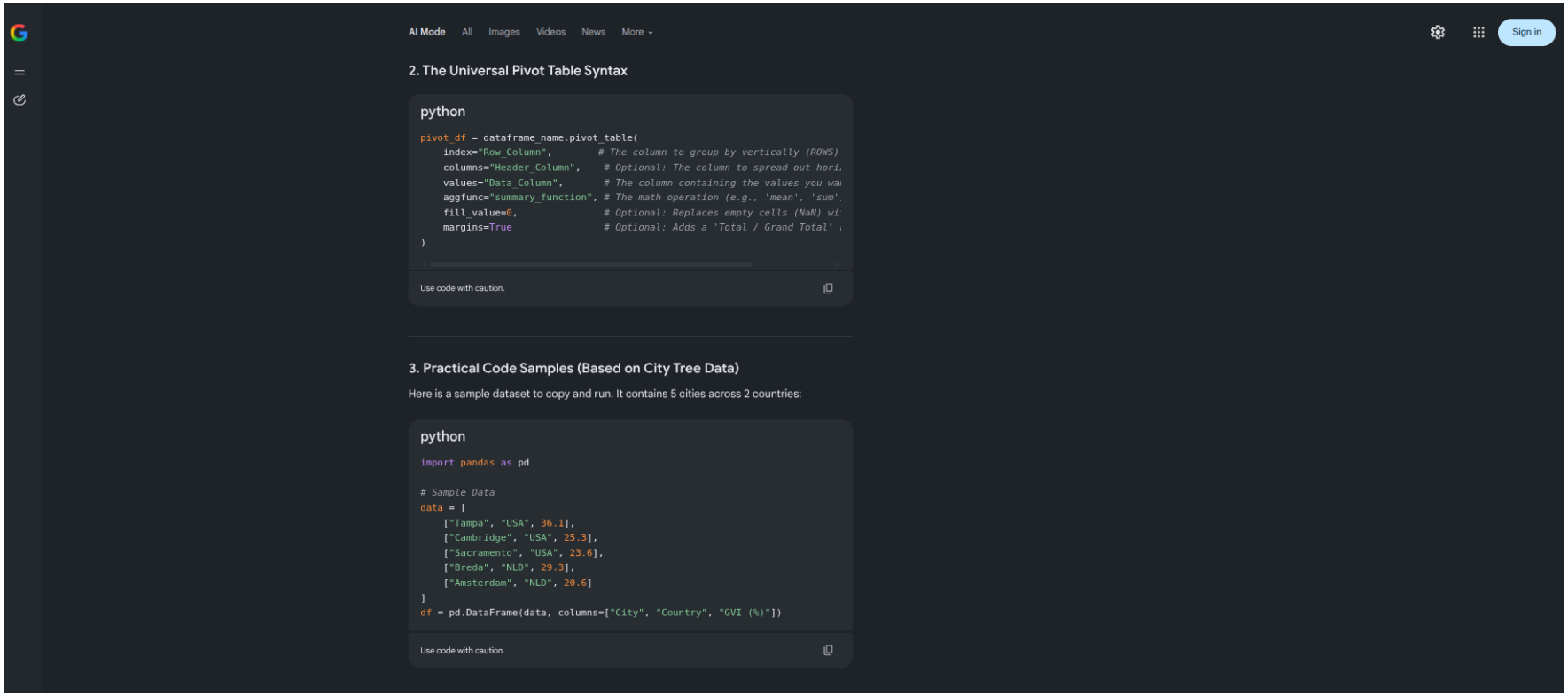
print("The count of number of cities of the maximum no of city's country is : ",max_city_count)
print("The country with maximum no of cities among top 30 is : ", top_country)
```

```
The count of number of cities of the maximum no of city's country is :  8
The country with maximum no of cities among top 30 is :  USA
```

Prompts taken for help using AI


google.com/search?q=hey&sourceid=chrome&ie=UTF-8&amc=1&gs_lcrp=EgZjaHJvbWUyBggAEEUYOTITCAEQABiDARixAxiABBjwBRieBjIQCAIQABixAxiABBjwBRieBjIQCAMQABix
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give me syntax of creating a pivot table from an existing dataframe
The syntax to create a pivot table from an existing DataFrame in pandas is:
python
pivot_table = dataframe_name.pivot_table(
values=None, # Column(s) to aggregate (numeric data)
index=None, # Column(s) to turn into ROWS
columns=None, # Column(s) to turn into COLUMNS
aggfunc='mean', # Math operation: 'sum', 'count', 'mean', 'max',
fill_value=None, # Replace empty cells/NaN (e.g., fill_value=0)
margins=False # Add a 'Total' row/column if set to True
)
Use code with caution.
Alternatively, you can call it directly using the pandas library module:
python
pivot_table = pd.pivot_table(dataframe_name, index="Row_Column", values="


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Basics needed to know to work with summarisation of data using pivot table in python give me with syntax and codes samples
To summarize data efficiently using a pivot table in Python, you need to understand four core concepts of the pandas library.
1. The Core Summary Vocabulary (aggfunc)
When summarizing data, you must tell pandas how to aggregate multiple rows into a single value. These are passed as strings inside quotes:
"mean" : Calculates the average value (default).
"sum" : Adds all numerical values together.
"count" : Counts the number of items or occurrences (works on text and numbers).
"max" / "min" : Finds the highest or lowest value in that group.
2. The Universal Pivot Table Syntax
python
pivot_df = dataframe_name.pivot_table(
index="Row_Column", # The column to group by vertically (ROWS)
columns="Header_Column", # Optional: The column to spread out hori
values="Data_Column", # The column containing the values you wa


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2. The Universal Pivot Table Syntax
python
pivot_df = dataframe_name.pivot_table(
index="Row_Column", # The column to group by vertically (ROWS)
columns="Header_Column", # Optional: The column to spread out hori
values="Data_Column", # The column containing the values you wa
aggfunc="summary_function", # The math operation (e.g., 'mean', 'sum'
fill_value=0, # Optional: Replaces empty cells (NaN) wi
margins=True # Optional: Adds a 'Total / Grand Total'
)
Use code with caution.
3. Practical Code Samples (Based on City Tree Data)
Here is a sample dataset to copy and run. It contains 5 cities across 2 countries:
python
import pandas as pd
Sample Data
data = [
["Tampa", "USA", 36.1],
["Cambridge", "USA", 25.3],
["Sacramento", "USA", 23.6],
["Breda", "NLD", 29.3],
["Amsterdam", "NLD", 20.6]
]
df = pd.DataFrame(data, columns=["City", "Country", "GVI (%)"])
Use code with caution.

In [ ]: